

ANWEASHA SAHA

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EDUCATION

University of Southern California, Los Angeles, USA

August 2023 – May 2025

Master of Science in Computer Science (Artificial Intelligence)

CGPA: 3.4/4

Relevant Courses: Machine Learning, Deep Learning and Its Applications, Applied Natural Language Processing, Web Technologies, Foundations of AI, Analysis of Algorithms, Advanced Computer Vision, Multimodal Probabilistic Learning

SRM Institute of Science and Technology, Kattankulathur, India

June 2019 – May 2023

B. Tech in Computer Science and Engineering

CGPA: 9.5/10

TECHNICAL SKILLS

- Languages: Python, SQL, C/C++, Java, JavaScript, HTML, RegEx
- Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Excel, Power BI, DAX, Tableau, Looker, IBM Cognos, SAS, Google Analytics
- Machine Learning & AI: Scikit-Learn, PyTorch, TensorFlow, Keras, EDA, Feature Engineering, Model Evaluation
- Databases & Data Engineering: MSSQL, PostgreSQL, dbt, ETL, Databricks, GCP, BigQuery, Spark, Snowflake, AWS
- Tools & Workflow Management: Git, Jira, Airflow
- Certifications & MOOCs: Machine Learning Specialisation by DeepLearning.AI, IBM Data Science Professional Certificate

WORK EXPERIENCE

USC University Advancement

Los Angeles, USA

Student Data Analyst (Business Intelligence Team)

March 2024 – May 2025

- Optimized SQL-based **production data pipelines** and donor-related queries, reducing execution time by **25%** while enhancing data quality and improving performance for enterprise-wide reporting systems.
- Developed scalable **Cognos reports** across departments, aligning with financial reporting standards, replacing outdated data sources with cleaned alternatives, and improving strategic visibility by **16%** for executive stakeholders.
- Designed **business intelligence dashboards** in Tableau with fiscal automation and consistent **KPIs**, enabling faster insights and improved collaboration across cross-functional teams.
- Automated reporting workflows using Python through scripted data handling and scheduled report distribution, **cutting manual effort by 57%** and accelerating reporting in agile sprints.

Decoders Labs

Remote, India

Python Developer Intern

December 2021 – March 2022

- Built robust **ETL pipelines** using Selenium to extract and structure grocery product and pricing data, automating ingestion into analytics-ready formats for the YourGrocer app.
- Created interactive dashboard visualizations using **Power BI** and **DAX**, highlighting item availability, customer ratings, and geographic trends to guide market expansion and service optimization.
- Analyzed user shopping patterns and marketing campaign response data, conducting **market basket analysis** and **A/B testing** to refine recommendation strategies—boosting relevance by **12%** and increasing transactions by **8%**.
- Conducted **root-cause analysis** and ad hoc data exploration to uncover regional demand shifts, pricing sensitivity, and marketing optimization opportunities, informing **product and outreach improvements**.

Technocolabs Softwares

Remote, India

Data Science Intern

August 2021 – September 2021

- Conducted analysis of 5 years' worth of **stock data** by integrating company news headlines with historical records, cleaning data, and applying rigorous **EDA** techniques.
- Developed a **Q-Learning** reinforcement learning model for stock price **forecasting**, benchmarked against Random Forest and SVM baselines to evaluate performance and adaptability.
- Built a user-friendly web app for daily stock predictions and investment recommendations, improving client decision-making by **7%**, and applied **feature engineering** techniques to enhance model performance using temporal and sentiment-based features.

PROJECTS

H&M Sales Dashboard – Retail & Customer Analytics Project

February 2025

- Built a **multi-layered Tableau dashboard** using H&M's Kaggle competition dataset with **31M+ transactions**, revealing patterns in customer demographics, purchase channels, and product types.
- Enabled interactive filtering and **customer segmentation** by product category, time period, and loyalty status to uncover high-performing segments and seasonal trends.
- Visualized key insights through a mix of line graphs, bar charts, sunburst diagrams, and customer distribution plots, offering a clear view of **purchase behavior** and revenue concentration.
- Highlighted shifts in **consumer preferences** across age groups, club membership status, and **seasonal color/pattern** trends — insights that could inform marketing segmentation and campaign targeting strategies.

Understanding the Relation between Noise and Bias in Annotated Datasets

USC ISI, USA

September 2023 – February 2024

- Analyzed annotation disagreements in NLP classification tasks using **statistical computing techniques** and data quality metrics to evaluate how models handle subjectivity in labels.
- Applied **quantitative methods** and **regression analysis** to compare Single vs Multi Ground Truth approaches, showing improved confidence in minority annotations with multi-annotator models.
- Created visual summaries to demonstrate the role of annotator consistency in improving model interpretability and reliability.
- **Co-authored** and submitted the paper *Don't Blame the Data, Blame the Model: Understanding Noise and Bias When Learning from Subjective Annotations* at the **EAACL 2024 Workshop UncertainNLP**; DOI: 10.48550/arXiv.2403.04085.